

1. (Spring 2024, #6) Find the sum of the series $\sum_{n=0}^{\infty} \frac{3(-1)^n}{2^{2n+1}}$.

$$= \sum_{n=0}^{\infty} \frac{3}{2} \cdot \frac{(-1)^n}{4^n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{3}{2}\right) \cdot \left(-\frac{1}{4}\right)^n.$$

$$\left|-\frac{1}{4}\right| < 1 \cdot \frac{3}{2} \cdot \frac{1}{1 - \left(-\frac{1}{4}\right)}$$

$$= \frac{6}{5}.$$

2. (Spring 2020, #9) Consider the sequence (a_n) defined recursively by

$$\begin{cases} a_0 = 1 \\ a_{n+1} = \frac{-2}{2a_n + 5}, \text{ for } n \geq 0 \end{cases}$$

You may freely use the following **fact**: (a_n) is always contained in $\left(-\frac{5}{2}, \frac{7}{2}\right)$.

- (a) Compute the values of a_0 , a_1 , and a_2 .

$$a_0 = 1, \quad a_1 = \frac{-2}{2 \cdot 1 + 5} = \frac{-2}{7}.$$

$$a_2 = \frac{-2}{2 \cdot \left(-\frac{2}{7}\right) + 5} = -\frac{14}{31}.$$

- (b) Use mathematical induction to determine whether (a_n) is increasing or decreasing.

$$\text{Note that } a_1 = \frac{-2}{7} = \frac{-14}{49} > \frac{-14}{31} = a_2.$$

$\{a_n\}$ is decreasing.

OR: consider the function

Suppose $a_n < a_{n-1} < 0$. Then

$$f(x) = \frac{-2}{2x+5}$$

$$f'(x) = \frac{4}{(2x+5)^2} > 0.$$

$$a_{n+1} = \frac{-2}{2a_n+5} < \frac{-2}{2a_{n-1}+5} = a_n. \Rightarrow f(x) \text{ is increasing.}$$

$$\Rightarrow a_{n+1} = f(a_n) < f(a_{n-1}) = a_n.$$

- (c) Show that (a_n) is convergent and determine the value of its limit. You must justify your findings.

since a_n is decreasing and $\{a_n\}$ is bounded.

Then $\{a_n\}$ is convergent by monotone convergence.

Let $a_n \rightarrow a$. Then

$$a = \frac{-2}{2a+5} \Rightarrow 2a^2 + 5a + 2 = 0 \Rightarrow (2a+1)(a+2) = 0.$$

$$\text{since } a \in \left(-\frac{5}{2}, \frac{7}{2}\right) \Rightarrow a = -\frac{1}{2} \text{ or } a = -2.$$

- (d) Determine the convergence of the series $\sum_{n=0}^{+\infty} a_n$. Justify your answer.

$$\begin{aligned} \sum_{n=0}^{\infty} a_n &= a_0 + \sum_{n=1}^{\infty} a_n. \\ &= a_0 - \sum_{n=1}^{\infty} (-a_n). \end{aligned}$$

where $-a_n > 0$.

$$\text{However } \lim_{n \rightarrow \infty} (-a_n) = \frac{1}{2} \neq 0. \Rightarrow \text{By divergence test}$$

The series is divergent.

At the limit, $|f'(a)| < 1$.

$\Rightarrow$ The limit $a = -\frac{1}{2}$. $|f'(-\frac{1}{2})| = 4$.

Note that $|f'(2)| = 4 > 1$.

$$|f(a_{n+2}) - f(a_{n+1})| < |f(a_{n+1}) - f(a_n)|$$

$$\Rightarrow |f(a_{n+2}) - f(a_{n+1})| < |a_{n+2} - a_{n+1}|$$

$$\Rightarrow \left| \frac{f(a_{n+2}) - f(a_{n+1})}{a_{n+2} - a_{n+1}} \right| < 1$$

$$\Rightarrow |f'(a)| < 1.$$

3. (Fall 2019, #6a) Suppose a particle is moving back and forth along the x -axis. The particle begins at the point $x = 0$. It then moves in the positive x -direction a distance of 1. The particle then moves in the negative x -direction a distance of $1/4$. The particle then moves in the positive x -direction a distance of $1/16$. The particle then moves in the negative x -direction a distance of $1/64$, and so on. In general, whenever the particle moves a distance h in one direction, it then moves in the opposite direction a distance of $h/4$. What position on the x -axis does the particle eventually approach?

$$\sum_{n=0}^{\infty} \left(\frac{1}{4}\right)^n \cdot (-1)^n$$

$$= \sum_{n=0}^{\infty} \left(-\frac{1}{4}\right)^n$$

$$\begin{aligned} & \left|-\frac{1}{4}\right| < 1 \\ & = \frac{1}{1 - \left(-\frac{1}{4}\right)} = \frac{4}{5} \end{aligned}$$

4. (Spring 2015, #6c,d)

(a) Find the limit if it exists of the sequence given by $u_n = \left(\frac{n+3}{n+1}\right)^n$.

$$\begin{aligned}\lim_{n \rightarrow \infty} u_n &= \lim_{n \rightarrow \infty} \left(1 + \frac{2}{n+1}\right)^n \\&= \lim_{n \rightarrow \infty} \frac{\left(1 + \frac{2}{n+1}\right)^{n+1}}{\left(1 + \frac{2}{n+1}\right)} \\&= \frac{e^2}{1} = e^2.\end{aligned}$$

(b) Is $\sum_0^{\infty} \left(\frac{n+3}{n+1}\right)^n$ convergent or divergent?

Divergent.

$$\text{since } \lim_{n \rightarrow \infty} \left(\frac{n+3}{n+1}\right)^n = e^2 \neq 0.$$

Then by the test of divergence,

the series diverges.

5. (Spring 2012, #8) Find the sum of the series $\sum_{n=2}^{\infty} \left[\frac{2}{n^2-1} + \left(\frac{2}{3} \right)^n \right]$.

$$= \sum_{n=2}^{\infty} \frac{2}{n^2-1} + \sum_{n=2}^{\infty} \left(\frac{2}{3} \right)^n$$

$$= \sum_{n=2}^{\infty} \frac{2}{(n+1)(n-1)} + \frac{\left(\frac{2}{3} \right)^2}{1 - \frac{2}{3}}$$

$$= \sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n+1} \right) + \frac{4}{3}$$

$$= \lim_{k \rightarrow \infty} \sum_{n=2}^k \left(\frac{1}{n-1} - \frac{1}{n+1} \right) + \frac{4}{3}$$

$$= \lim_{k \rightarrow \infty} \left(1 - \cancel{\frac{1}{3}} + \cancel{\frac{1}{2}} - \cancel{\frac{1}{4}} + \cancel{\frac{1}{3}} - \cancel{\frac{1}{5}} + \dots + \cancel{\frac{1}{k-1}} - \cancel{\frac{1}{k+1}} \right) + \frac{4}{3}.$$

$$= \lim_{k \rightarrow \infty} \left(1 + \frac{1}{2} - \frac{1}{k} - \frac{1}{k+1} \right) + \frac{4}{3}$$

$$= \frac{3}{2} + \frac{4}{3} = \frac{17}{6}.$$